

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408583
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Guo; Jianpeng

String trimmer and trimming head thereof

Abstract

A string trimmer includes a trimming head, a driving device, and an operating device. The trimming head includes a spool, a head housing formed with an accommodating space, and a transmission member mounted to the spool that can move freely between the head housing and the spool in the direction of a straight line perpendicular to the direction of the rotation axis. The trimming line has an effective portion extending from the accommodating space. When the length of the effective portion of the trimming line is larger than a preset value, the transmission member automatically releases the head housing. When the length of the effective portion of the trimming line is less than the preset value, the transmission member locks the head housing.

Inventors:	Guo; Jianpeng (Nanjing, CN)
Applicant:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Family ID:	73749680
Assignee:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Appl. No.:	17/551282
Filed:	December 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220104431 A1	Apr. 07, 2022

Foreign Application Priority Data

CN	201910520911.X	Jun. 17, 2019
----	----------------	---------------

Related U.S. Application Data

Publication Classification**Int. Cl.:** A01D34/416 (20060101); A01D101/00 (20060101)**U.S. Cl.:****CPC** A01D34/4162 (20130101); A01D34/4166 (20130101); A01D2101/00 (20130101)**Field of Classification Search****CPC:** A01D (34/4162); A01D (34/4166); A01D (34/00); A01D (34/4161); A01D (34/416); A01D (34/4163); A01D (34/4165); A01D (34/4167); A01D (34/4168); A01D (69/06); A01D (75/20)**USPC:** 30/276

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4419822	12/1982	Harris	N/A	N/A
4888871	12/1988	Engelbrecht	30/276	A01D 34/4162
7797839	12/2009	Proulx	N/A	N/A
2009/0260237	12/2008	Alliss	N/A	N/A
2016/0183452	12/2015	Kullberg	N/A	N/A
2016/0249529	12/2015	Ma	30/276	A01D 34/4161
2017/0347523	12/2016	Alliss	N/A	N/A
2018/0098492	12/2017	Guo	N/A	N/A
2020/0128728	12/2019	Guo	N/A	N/A
2021/0378173	12/2020	Cholst	N/A	N/A
2022/0104430	12/2021	Guo	N/A	N/A
2023/0397527	12/2022	Guo	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105815028	12/2015	CN	N/A
109379973	12/2016	CN	N/A
108135129	12/2017	CN	N/A
109691287	12/2018	CN	N/A
111096141	12/2022	CN	N/A
2798934	12/2013	EP	N/A
2016173657	12/2015	WO	N/A

OTHER PUBLICATIONS

ISA/CN, International Search Report issued on PCT application No. PCT/CN2020/096353, dated Sep. 21, 2020, 4 pages. cited by applicant

ISA/CN, English translation of International Search Report issued on PCT application No. PCT/CN2020/096353, dated Sep. 21, 2020, 2 pages. cited by applicant

EPO, extended European search report issued on European patent application No. 20826472.1, dated Jun. 20, 2022, 7 pages. cited by applicant
CIPO, office action issued on Canadian patent application No. 3,144,111, dated Mar. 15, 2023, 17 pages. cited by applicant
Office Action from Canadian application No. 3,144,111, dated Mar. 20, 2024, 8 pp. cited by applicant
Non-final Office Action from U.S. Appl. No. 17/551,225, dated Mar. 26, 2025, 14 pp. cited by applicant
International Search Report and Written Opinion of application No. PCT/CN20/96353, dated Sep. 21, 2020, 6 pp. cited by applicant
Written Opinion of international application No. PCT/CN20/96354, dated Dec. 21, 2017, 4 pp. cited by applicant

Primary Examiner: Eiseman; Adam J

Assistant Examiner: Cornett; Robert D

Attorney, Agent or Firm: Greenberg Traurig, LLP

Background/Summary

RELATED APPLICATION INFORMATION (1) This application is a continuation of International Application Number PCT/CN2020/096353, filed on Jun. 16, 2020, through which this application also claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. CN 201910520911.X, filed on Jun. 17, 2019, which applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a garden tool, for example, a string trimmer and a trimming head thereof.

BACKGROUND

(2) As a garden tool, a string trimmer can be used for trimming grass in lawn of villas and gardens. At present, the string trimmer on the market mainly use tapping or a switch to control a spool and a head housing of the trimming head to produce a speed difference to feed a line. Currently users cannot accurately feed the length of the line according to actual needs. In fact, none of the known trimming heads can realize automatic feed in the true sense.

(3) Generally speaking, the trimming head with an automatic feeding function causes a trimming line to be released by causing a relative rotation between the head housing and the spool. There are many ways to make the head housing and the spool rotate relative to each other. In the related art, the automatic feeding of the string trimmer generally requires the outside to input an instruction signal to the trimming head, and the string trimmer must be controlled to feed the trimming line according to the human judgment of an operator. That is, when the operator observes that the length of the trimming line changes and the trimming efficiency is reduced, the string trimmer is manually controlled to feed. This includes the control of the string trimmer to accelerate or decelerate through the feeding switch for the speed difference between the head housing and the spool, or through tapping the trimming head to cause the speed difference between the head housing and the spool to achieve a line-feeding. For this type of string trimmer, when the line is automatically feeding, the trimming head will produce a large speed difference when it is driven, and generally will produce a speed difference of more than 30%, so as to achieve a speed difference between the

head housing and the spool. As for some string trimmer with automatic feeding function in related technologies, they are set so that when the string trimmer is turned on or off, there will also be a speed difference between the head housing and the spool to achieve line feeding, which is not effective and reliable line feeding based on actual needs. Under the premise that the trimming line itself has an effective cutting length, this kind of line feeding requires the user to cut off the excess line, which actually causes a certain degree of waste of resources, and it will cause some unnecessary troubles to the user in the actual operation process, and seriously affect the convenience of the user's operation.

(4) There are also some documents that disclose that a sensor is installed inside the trimming head to sense the length of the trimming line and transmit the signal to a control portion to control the realization of the acceleration or deceleration of the trimming head. It also requires the trimming head to have a large speed difference, and the speed difference is generally required to be more than 30%, and the string trimmer that senses the signal and controls the feeding through the sensor is not reflected in the products of the related technology. It is obtained through multi-party demonstration that in the actual feeding process, there will generally be situations where the line cannot be put out or the line is too long. The reliability of this type of string trimmer is extremely poor, and it cannot adapt to a more complicated working condition when the string trimmer is working.

SUMMARY

(5) In one example, a string trimmer includes: a trimming head; and a driving device including a motor for driving the trimming head to rotate around a rotation axis; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member mounted to the spool, wherein the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; wherein the head housing is formed with a mating portion to mate with the driving portion; wherein the mating portion is formed with a mating surface for contacting the driving portion to apply a reacting force to the driving portion and a return surface for contacting the driving portion to apply a reacting force to the driving portion; when the driving portion mates with the mating surface, the trimming head is in a working state; when the driving portion mates with the return surface, the trimming head is in an autonomous line-feeding state; when the trimming head is in the autonomous line-feeding state and the driving portion mates with the return surface, the return surface has a projection line in a plane perpendicular to the rotation axis, and the projection line includes a first extreme position and a second extreme position that can push the transmission member to return to the mating surface; the string trimmer further includes a first circumference centered on the rotation axis and passing through the first extreme position, and a second circumference centered on the rotation axis and passing through the second extreme position; the angle formed by a tangent line of the first circumference at a first extreme position and a tangent line of the second circumference at the second extreme position is less than or equal to 45° ; relative to the spool, the transmission member further has a first position that prevents a relative rotation between the spool and the head housing and a second position that allows the relative rotation between the spool and the head housing; wherein, when the transmission member is in the first position, the trimming head is in a trimming state, and the motor has a first rotation speed; when the transmission member is in the second position, the trimming head is in the autonomous line-feeding state, the motor has a second rotation speed, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.15, and the relative rotation is generated between the spool and the head housing in order to feed the trimming line.

(6) In one example, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or

equal to 0.1.

(7) In one example, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.15.

(8) In one example, the trimming line has an effective length extending from the accommodating space, and the trimming line has a preset value within an effective length range; when the effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(9) In one example, the transmission member includes the driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with the mating portion for mating with the driving portion; the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis; in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(10) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(11) In one example, a central angle β formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(12) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through the first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a direction of a straight line perpendicular to the rotation axis.

(13) In one example, when the trimming head is driven by the motor to rotate around the rotation axis, the transmission member has a friction force in a straight line perpendicular to the rotation axis relative to the spool, and the friction force is in the opposite direction of the centrifugal force generated by the transmission member.

(14) In one example, the spool is further formed with or connected with a receiving groove, and a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(15) In one example, a string trimmer includes: a trimming head; and a driving device including a motor for driving the trimming head to rotate around a rotation axis; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a linkage device for linking the spool and head housing; wherein the linkage device includes: a transmission member having a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the relative rotation between the spool and the head housing relative to the spool; wherein when the transmission member is in the first position, the trimming head is in a trimming state, and the motor has a first rotation speed; when the transmission member is in the second position, the trimming head is in the autonomous line-feeding state, the motor has a second rotation speed, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.05; the relative rotation is

generated between the spool and the head housing to feed the trimming line.

(16) In one example, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.1.

(17) In one example, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.15.

(18) In one example, the trimming line has an effective length extending from the accommodating space, and the trimming line has a preset value within an effective length range; when the effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(19) In one example, the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion for mating with the driving portion; the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis; in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(20) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(21) In one example, a central angle β formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(22) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through the first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a direction of a straight line perpendicular to the rotation axis.

(23) In one example, when the trimming head is driven by the motor to rotate around the rotation axis, the transmission member has a friction force in a straight line perpendicular to the rotation axis relative to the spool, and the friction force is in the opposite direction of the centrifugal force generated by the transmission member.

(24) In one example, the spool is further formed with or connected to a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against a bottom of the receiving groove, and the other end abuts against the transmission member.

(25) In one example, a string trimmer includes: a trimming head; and a driving device including a motor for driving the trimming head to rotate around a rotation axis; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member having a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the relative rotation between the spool and the head housing relative to the spool; wherein the trimming head is in a working state and the transmission member rotates around the rotation axis synchronously with the spool when the transmission member is in the first position; the trimming head is in the autonomous line-feeding state and the relative rotation is generated between the spool and the head

housing to feed the trimming line when the transmission member is in the second position; wherein the center of mass of the transmission member deviates from the rotation axis; when the trimming head is driven to rotate by the motor, the transmission member is subjected to a centrifugal force which cause a moving tendency in a direction from the first position to the second position; when the head housing rotates around the rotation axis, the head housing also applies a resistance to the transmission member against the centrifugal force to prevent the transmission member from moving from the first position to the second position; when the trimming head is in a trimming state and the length of an effective portion of the trimming line extending beyond the head housing is less than a preset value, the resistance applied by the head housing to the transmission member is reduced so that the transmission member moves from the first position to the second position under the centrifugal force.

(26) In one example, when an effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(27) In one example, the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion to mate with the driving portion; the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis.

(28) In one example, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(29) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(30) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide; the limiting member and the limiting portion include at least two contact surfaces in a straight line perpendicular to the rotation axis.

(31) In one example, when the trimming head is driven by the motor to rotate around the rotation axis, the transmission member has a friction force in a straight line perpendicular to the first rotation axis relative to the spool, and the friction force is in the opposite direction of the centrifugal force generated by the transmission member.

(32) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(33) In one example, a trimming head is used for a string trimmer and is capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing surrounding to form an accommodating space; a transmission member having a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the relative rotation between the spool and the head housing relative to the spool; wherein the transmission member rotates around the rotation axis synchronously with the spool and the trimming head is in a trimming state when the transmission member is in the first position; the relative rotation is generated between the spool and the head housing to feed the trimming line when the transmission member is in the second position; the center of mass of the transmission member deviates from the rotation axis; when the trimming head is rotating, the transmission member is subjected to a

centrifugal force which cause a moving tendency in a direction from the first position to the second position; when the head housing rotates around the rotation axis, the head housing also applies a resistance to the transmission member against the centrifugal force to prevent the transmission member from moving from the first position to the second position; when the trimming head is in the trimming state and the length of an effective portion of the trimming line extending beyond the head housing is less than a preset value, the resistance applied by the head housing to the transmission member is reduced so that the transmission member moves from the first position to the second position under the centrifugal force.

(34) In one example, the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion for mating with the driving portion; the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis; in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(35) In one example, a string trimmer includes: a trimming head for trimming grass; and a driving device including a motor for driving the trimming head to rotate around a rotation axis; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a linkage device for realizing power transmission between the spool and the head housing; wherein the string trimmer has a trimming state, and the linkage device drives the spool or the head housing to rotate around the rotation axis and in a first rotation direction when the string trimmer is in a working state; wherein the linkage device includes: a transmission member having a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the relative rotation between the spool and the head housing relative to the spool; the trimming head also has an automatic winding state and an autonomous line-feeding state; when the trimming head is in the automatic winding state, the motor drives the spool to rotate relative to the head housing in a second rotation direction so that the trimming line is wound to the spool; when the trimming head is in the autonomous line-feeding state, the transmission member moves to the second position, and the spool rotates relative to the head housing along the first rotating direction to release the trimming line; wherein the transmission member is subjected to a centrifugal force which cause a moving tendency in a direction from the first position to the second position when the trimming head is driven to rotate by the motor; when the head housing rotates around the rotation axis, the head housing also applies a resistance to the transmission member against the centrifugal force to prevent the transmission member from moving from the first position to the second position; when the trimming head is in the working state and the length of an effective portion of the trimming line extending beyond the head housing is less than a preset value, the resistance applied by the head housing to the transmission member is reduced so that the transmission member moves from the first position to the second position under the centrifugal force.

(36) In one example, a damping device for damping at least one of the spool or the head housing to make the string trimmer in the automatic winding state is further included.

(37) In one example, the damping device includes: a damping member for preventing the spool from rotating in a first direction; wherein in the automatic winding state, the head housing rotates in the first direction.

(38) In one example, the damping device includes: a damping member for preventing the head housing from rotating in a second direction; wherein in the automatic winding state, the spool rotates in the second direction.

(39) In one example, the transmission member includes a driving portion for driving the head

housing to rotate synchronously with the spool; the head housing is formed with a mating portion for mating with the driving portion.

(40) In one example, the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis;

(41) In one example, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(42) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects a mating surface and is away from the rotation axis, and the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(43) In one example, a central angle α formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(44) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a straight line perpendicular to the rotation axis.

(45) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(46) In one example, a trimming head is used for string trimmer and capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member mounted to the spool, wherein the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; wherein the head housing is formed with a mating portion to mate with the driving portion; wherein the mating portion is formed with a mating surface for contacting with the driving portion to apply a reacting force to the driving portion, the mating surface extends in a first plane, and the first plane is substantially parallel to the rotation axis; and the distance between the rotation axis and the first plane is greater than zero.

(47) In one example, the distance between the rotation axis and the first plane is greater than 0 and less than or equal to 20 mm.

(48) In one example, the mating surface includes an effective surface capable of applying the reaction force, the effective surface having a projection line in a plane perpendicular to the rotation axis, the projection line includes a first extreme position and a second extreme position, and the central angle between the first extreme position and the second extreme position and the axis of the rotation axis is greater than 0° and less than or equal to 45° .

(49) In one example, the central angle between the first extreme position and the second extreme position and the axis of the rotation axis is greater than or equal to 3° and less than or equal to 40° .

(50) In one example, relative to the spool, the transmission member has a first position that prevents a relative rotation between the spool and the head housing and a second position that allows the relative rotation between the spool and the head housing; the trimming line has an effective length extending from the accommodating space, and the trimming line has a preset value within an effective length range; when the effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(51) In one example, the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis.

(52) In one example, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(53) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(54) In one example, a central angle α formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(55) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a straight line perpendicular to the rotation axis.

(56) In one example, a trimming head is used for string trimmer and is capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member mounted to the spool, wherein the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; wherein the head housing is formed with a mating portion to mate with the driving portion; wherein the mating portion is formed with a mating surface for contacting the driving portion to apply a reaction force to the driving portion, the mating surface includes an effective surface capable of applying the reaction force, the effective surface has a projection line in a plane perpendicular to the rotation axis, the projection line includes a first extreme position and a second extreme position, and the central angle between the first extreme position and the second extreme position and the axis of the rotation axis is greater than 0° and less than or equal to 45° .

(57) In one example, the central angle between the first extreme position and the second extreme position and the axis of the rotation axis is greater than or equal to 3° and less than or equal to 40° .

(58) In one example, relative to the spool, the transmission member has a first position that prevents a relative rotation between the spool and the head housing and a second position that allows the relative rotation between the spool and the head housing; the trimming line has an effective length extending from the accommodating space, and the trimming line has a preset value within an effective length range; when the effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(59) In one example, the mating portion includes first teeth distributed around the rotation axis; the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis;

(60) In one example, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

(61) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to

rotate around the rotation axis.

(62) In one example, a central angle β formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(63) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed to provide a sliding rail for the transmission member to slides, and the limiting member and the limiting portion include at least two contact surfaces in the direction of a straight line perpendicular to the rotation axis.

(64) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(65) In one example, a trimming head is used for string trimmer and is capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member mounted to the spool, wherein the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; wherein the head housing is formed with a mating portion to mate with the driving portion; wherein the mating portion is formed with a mating surface for contacting with the driving portion to apply a reacting force to the driving portion; the mating surface has a projection line in a plane perpendicular to a rotation axis; the projected line includes a first extreme position and a second extreme position at which a reacting force can be applied to the driving portion, the mating surface includes an effective action surface that can apply the reacting force, and the effective action surface is disposed between the first extreme position and the second extreme position.

(66) In one example, a trimming head is used for string trimmer and is capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a transmission member mounted to the spool, wherein the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; wherein the head housing is formed with a mating portion to mate with the driving portion; wherein the mating portion is formed with a mating surface for contacting the driving portion to apply a reacting force to the driving portion, and a return surface for contacting the driving portion to apply a reacting force to the driving portion; when the driving portion mates with the mating surface, the trimming head is in a working state; when the driving portion mates with the return surface, the trimming head is in an autonomous line-feeding state; when the trimming head is in the autonomous line-feeding state and the driving portion mates with the return surface, the return surface has a projection line in a plane perpendicular to the rotation axis, and the projection line includes a first extreme position and a second extreme position that can push the transmission member to return to mate with the mating surface; the string trimmer further includes a first circumference centered on the rotation axis and passing through the first extreme position, and a second circumference centered on the rotation axis and passing through the second extreme position; the angle formed by a tangent line of the first circumference at a first extreme position and a tangent line of the second circumference at the second extreme position is less than or equal to 45° .

(67) In one example, the angle formed by the tangent of the first circumference at the first extreme position and the tangent of the second circumference at the second extreme position is less than or equal to 30° .

(68) In one example, the transmission member has a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the

relative rotation between the spool and the head housing relative to the spool;

(69) In one example, the trimming line has an effective length extending from the accommodating space, and the trimming line has a preset value within an effective length range; when the effective length of the trimming line is greater than the preset value, the transmission member is in the first position; when the effective length of the trimming line is less than the preset value, the transmission member is in the second position.

(70) In one example, when the driving portion moves to the first extreme position and the second extreme position, the trimming line has a reacting force acting on the head housing, and the return surface has a first component force that pushes the transmission member to reset and a second force component that is opposite to the reacting force applied to the head housing by the trimming line.

(71) In one example, the mating portion includes first teeth distributed around the rotation axis; the first teeth are formed or connected with the mating surface;

(72) In one example, the head housing is formed with or connected with second teeth for resetting the transmission member, each of the second teeth is formed with or connected with a return surface; the first teeth and the second teeth are staggered around the rotation axis.

(73) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(74) In one example, a central angle β formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(75) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a straight line perpendicular to the rotation axis.

(76) In one example, when the trimming head is driven by the motor to rotate around the rotation axis, the transmission member has a friction force in a straight line perpendicular to the rotation axis relative to the spool, and the friction force is in the opposite direction of the centrifugal force generated by the transmission member.

(77) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(78) In one example, a trimming head is used for string trimmer is capable of being driven to rotate around a rotation axis to achieve grass trimming; wherein the trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a linkage device for realizing power transmission between the spool and the head housing; wherein the head housing is formed with an outer threading aperture capable of inserting the trimming line into the spool, and the spool is formed with a coupling portion capable of allowing the trimming line inserted into the head housing from the outer threading aperture to be connected to the spool; the linkage device includes a transmission member mounted to the spool, and the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion to mate with the driving portion; when the trimming head rotates around the rotation axis, and when the transmission member rotates with the spool around the rotation axis, the transmission member generates a centrifugal force along a straight line, and the mating portion and the driving portion contacts and applies a force to the driving portion; the force is decomposed into a first component force in a direction perpendicular to the straight line and a second component force along the direction of the straight line and opposite to the direction of the

centrifugal force to prevent the transmission member from moving along the direction of the straight line.

(79) In one example, the spool is formed with an inner threading aperture, and when the outer threading aperture and the inner threading aperture are located in a same radial direction, the trimming line can pass through the outer threading aperture and the inner threading aperture in sequence.

(80) In one example, the coupling portion is a threading channel, and the transmission channel penetrates the spool and connects any two inner threading apertures on the spool.

(81) In one example, the threading channel extends along a curve.

(82) In one example, the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion to mate with the driving portion.

(83) In one example, the mating portion includes first teeth distributed around the rotation axis;

(84) In one example, the head housing is formed with or connected with second teeth for resetting the transmission member; the first teeth and the second teeth are staggered around the rotation axis;

(85) In one example, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a protrusion facing the rotation axis and the protrusion is gradually changing.

(86) In one example, the tooth surface includes a first section and a second section; the first section and the tooth surface of the first teeth away from the rotation axis form a section for the transmission member to move radially along a direction perpendicular to the rotation axis; the second section and the tooth surface of the first teeth away from the rotation axis form a guiding channel for the transmission member to rotate around the rotation axis.

(87) In one example, a central angle α formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(88) In one example, a limiting portion is formed on the spool to limit the rotation of the transmission member around the rotation axis, and the limiting portion is symmetrically arranged with respect to a symmetry plane passing through a first straight line; the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting member and the limiting portion include at least two contact surfaces in a straight line perpendicular to the rotation axis.

(89) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove, one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(90) In one example, a trimming head includes: a spool for winding a trimming line; a head housing formed with an accommodating space, wherein the spool is at least partially accommodated in the accommodating space; and a linkage device for realizing power transmission between the spool and the head housing; wherein the head housing is formed with an outer threading aperture capable of inserting the trimming line into the spool, and the spool is formed with a coupling portion capable of allowing the trimming line inserted into the head housing from the outer threading aperture to be connected to the spool; wherein the linkage device includes: a transmission member has a first position that prevents a relative rotation between the spool and the head housing relative to the spool and a second position that allows the relative rotation between the spool and the head housing relative to the spool; wherein when the transmission member is in the first position, the trimming head is in a working state, and the transmission member rotates around the rotation axis synchronously with the spool; when the transmission member is in the second position, the trimming head is in an autonomous line-feeding state, the relative rotation is generated between the spool and the head housing to feed the trimming line.

(91) In one example, a transmission member mounted to the spool, and the transmission member includes a driving portion for driving the head housing to rotate synchronously with the spool; the head housing is formed with a mating portion to mate with the driving portion; when the trimming

head rotates around a rotation axis, and when the transmission member rotates with the spool around the rotation axis, the transmission member generates a centrifugal force along a straight line, and the mating portion and the driving portion contacts and applies a force to the driving portion; the force is decomposed into a first component force in a direction perpendicular to the straight line and a second component force along the direction of the straight line and opposite to the direction of the centrifugal force to prevent the transmission member from moving along the direction of the straight line.

(92) In one example, the spool is formed with an inner threading aperture, and when the outer threading aperture and the inner threading aperture are located in a same radial direction, the trimming line can pass through the outer threading aperture and the inner threading aperture in sequence.

(93) In one example, the coupling portion is a threading channel, and the transmission channel penetrates the spool and connects any two inner threading apertures on the spool.

(94) In one example, the threading channel extends along a curve.

(95) In one example, when the trimming head is driven by the motor to rotate around the rotation axis, the transmission member has a friction force in a straight line perpendicular to the rotation axis relative to the spool, and the friction force is in the opposite direction of the centrifugal force generated by the transmission member.

(96) In one example, the mating portion includes first teeth distributed around the rotation axis; the first teeth are formed or connected with the mating surface;

(97) In one example, the head housing is formed with or connected with second teeth for resetting the transmission member, each of the second teeth is formed with or connected with a return surface; the first teeth and the second teeth are staggered around the rotation axis.

(98) In one example, the return surface includes a first section and a second section, and the first section and the driving portion substantially do not generate an interaction force; each of the first teeth includes a tooth surface that intersects the mating surface and is away from the rotation axis; the second section and the tooth surface form a guiding channel for the transmission member to rotate around the rotation axis.

(99) In one example, a central angle β formed by a tooth tip and a tooth bottom of the second teeth and the axis of the rotation axis is greater than or equal to 30° and less than or equal to 50° .

(100) In one example, the spool is further formed with or connected with a receiving groove, a biasing element is arranged in the receiving groove; one end of the biasing element abuts against the bottom of the receiving groove, and the other end abuts against the transmission member.

(101) In one example, the automatically feeding of the trimming line of the trimming head itself when the trimming line is shortened to the preset length is realized to meet the demand for the length of the trimming line during trimming by setting the transmission member mated with the driving device and through purely mechanical cooperation. There is no need to manually determine the conditions of the feeding, and the reliability and the using convenience of the line-feeding of the string trimmer is improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a string trimmer;

(2) FIG. 2 is a section view of a trimming head for the string trimmer of FIG. 1;

(3) FIG. 3 is an explosion view of the trimming head for the string trimmer of FIG. 1;

(4) FIG. 4 is an explosion view of the trimming head for the string trimmer on another view;

(5) FIG. 5 is a section view of the spool of the string trimmer in FIG. 3;

(6) FIG. 6 is a perspective view of a spool and a transmission part of the string trimmer of FIG. 3;

- (7) FIG. 7 is a top view of a head housing, the spool and the transmission part of the string trimmer of FIG. 3;
- (8) FIG. 8 is a section view of the head housing, the spool and the transmission part of the string trimmer of FIG. 3;
- (9) FIG. 9 is a top view of the head housing of the string trimmer of FIG. 3;
- (10) FIG. 10 is a force analysis diagram for any point on a first teeth of the working head of the string trimmer of FIG. 8;
- (11) FIG. 11 is a section view of the working head, the spool and the transmission part of the string trimmer of FIG. 3 when the transmission part is between a first inner tooth and a second inner tooth;
- (12) FIG. 12 is a section view of the trimming head, the spool and the transmission part of the string trimmer of FIG. 3 when the transmission part is pushed back by the second inner tooth; and
- (13) FIG. 13 is a section view of the head housing, the spool and the transmission member of the string trimmer in FIG. 12 when the transmission member is pushed back by the second teeth; and in the view, part of the first teeth and the plurality of second teeth are deleted.
- DETAILED DESCRIPTION

- (14) A string trimmer **100** shown in FIG. 1 includes a trimming head **11**, a driving device **12**, an operating device **13**, and a connecting device **14**.
- (15) As shown in FIG. 1 to FIG. 2, the trimming head **11** is used to install a trimming line **15** for realizing the trimming function. The driving device **12** is used to provide a rotational power to the trimming head **11**. The driving device **12** includes a motor **122** and a first housing **121**. The motor **122** is arranged in the first housing **121**. The motor **122** drives the trimming head **11** to rotate by a rotation axis **101**. The operating device **13** includes a handle **131**, an auxiliary handle **132**, a main switch **133** and a second housing **134**. The handle **131** and the auxiliary handle **132** are used for the users to hold by two hands separately, so that the string trimmer **100** can be operated more stably. The main switch **133** can be positioned on the handle **131**, and the user can directly operate the main switch **133** to control the string trimmer **100** to trim grass when holding the handle **131**. The second housing **134** is for forming a coupling portion that is combined with the power supply device. For example, the coupling portion can be combined with a battery pack to supply power to the string trimmer **100**. In this example, the handle **131** and the second housing **134** are formed separately. In some other examples, the handle **131** can also be integrally formed with the second housing **134**. The connecting device **14** includes a connecting rod for connecting the first housing **121** and the second housing **134**. The auxiliary handle **132** is also installed to the connecting rod, and the auxiliary handle **132** is also located between the first housing **121** and the second housing **134**. The string trimmer **100** further includes a guard **16** which is at least partially surrounding the trimming head **11**, which in turn prevent the debris from flying to the direction where the user or operator stands when the string trimmer **11** is trimming the grass.
- (16) In order to facilitate the description of the technical solution of the present disclosure, an upper side and a lower side are defined as shown in FIG. 1, wherein the driving device **12** is arranged on the upper side of the trimming head **11**, and the trimming head **11** is arranged on the lower side of the driving device **12**.
- (17) The trimming head **11** also includes a head housing **111**, a spool **112** and a linkage device **113**. The head housing **111** is formed with an accommodating space **114** around the rotation axis **101** and the accommodating space **114** can accommodate at least a part of the spool **112**. The side of the head housing **111** forming the accommodating space **114** is the inner side of the head housing **111**, and the side of the head housing **111** opposite to the inner side is outside.
- (18) As shown in FIG. 1 to FIG. 4, the spool **112** is disposed in the accommodating space **114**. That is, the spool **112** is disposed on the inner side of the head housing **111**. The spool **112** is used for winding the trimming line **15**, and the trimming line **15** is used for trimming grass. The head housing **111** also forms an outer threading aperture **111a** for the trimming line **15** to extend to the

outside of the head housing **111**, and the amount of the outer threading apertures **111a** is two. The spool **112** also forms an inner threading aperture **112a**. When the inner threading aperture **112a** and the outer threading aperture **111a** are located in the same radial direction, the trimming line **15** can pass through the outer threading aperture **111a** and the inner threading aperture **112a** in sequence. Exemplarily, the spool **112** is also formed with a threading channel **112d** which connects two inner threading apertures **112a** and allows the trimming line **15** to pass through. In fact, the threading channel **112d** connects any two inner threading apertures on the spool, and the threading channel is arranged to extend along a curve. When the users need to supplement feed the trimming line **15**, the users can insert the trimming line **15** through the outer threading aperture **111a** into the accommodating space **114**, then pass the threading channel **112d** through the inner threading aperture **112a**, and to the outside of the head housing **111** from the outer threading aperture **111a** on the opposite side. When the trimming line **15** needs to be wound around the spool **112**, users do not need to open the head housing **111**. It is possible to directly extend the trimming line **15** into the head housing **111** and then wind the trimming line **15** on the spool **112** through the relative rotation of the spool **112** and the head housing **111**. Such type of trimming head **11** is usually called an external inserted winding head.

(19) The part of the trimming line **15** extending through the outer threading aperture **111a** to the outside of the head housing **111** is defined as an effective portion **15a** of the trimming line **15**, and the effective portion **15a** of the trimming line **15** trims weeds by high-speed rotation. In order to trim weeds within the expected area, the length of the effective portion **15a** of the trimming line **15** should reach a preset value. That is, the length of the effective portion **15a** of the trimming line **15** should be greater than or equal to the preset value. When the length of the effective portion **15a** of the trimming line **15** is less than the preset value, the cutting efficiency of the trimming line **15** may be reduced because the effective portion **15a** of the trimming line **15** is relatively short. In order to keep the length of the effective portion **15a** of the trimming line **15** to be greater than or equal to the length range of the preset value, the string trimmer **100** in this example can realize it that the linkage device **113** controls the trimming line **15** to be autonomously fed to a length range greater than or equal to the preset value when the length of the effective portion **15a** of the trimming line **15** is lower than the preset value. In order to prevent the extension length of the trimming line **15** from being too long and causing interference with the guard **16**, the effective portion **15a** of the trimming line **15** is actually less than or equal to a limit value. Within this limit, on one hand, the trimming line **15** can effectively mowing the grass and maintain a high mowing efficiency; on the other hand, the trimming line **15** will not interfere with the guard **16**, so that mowing can be continued.

(20) The linkage device **113** is used to control whether the trimming head **11** is feeding or not. The linkage device **113** has a first equilibrium state that makes the trimming head **11** to be unable to feed. At this time, the trimming head **11** is in a working state. The linkage device **113** also has a second equilibrium state in which the trimming head **11** can feed the trimming line, and the trimming head **11** is in a autonomously feeding state at this time. In this example, the linkage device **113** can autonomously identify a effective length of the trimming line **15** and can autonomously adjust the trimming head **11** to enter the first equilibrium state or the second equilibrium state according to the length of the effective portion **15a** of the trimming line **15**, that is to adjust the trimming head **11** to autonomously enter the working state or the autonomously feeding state. It needs to be emphasized that, for the linkage device **113**, the autonomous reorganization and adjustment of the working mode of the trimming head **11** is based on the linkage device **113** itself, and there is no need to accept an input from an outside or an indication signal for feeding the line or trimming. The indication signal mentioned here includes a signal that causes the linkage device **113** to switch modes due to an external force, electrical signal, magnetic force, or other force generated by the outside on the trimming head **11**. In some common indication signals, it generally includes: tapping the shell of the trimming head **11** to make the spool **112** and the head

housing **111** produce a speed difference to feed the trimming line; adjusting the speed through a switch to make the spool **112** and the head housing **111** to produce a speed difference to feed the trimming line; inducting the current change through a sensor induces in the motor to cause a speed difference between the spool **112** and the head housing **111** to feed the trimming line. In addition, the rotation speed of the trimming head **11** in this example is constant, and there is no need to change the rotation speed of motor **122** to make the spool **112** to have a speed change to obtain the speed difference between the head housing **111** and the spool **112** to perform line feeding.

(21) Exemplarily, when the linkage device **113** is in the first equilibrium state, the spool **112** and the head housing **111** are relatively static. In this way, the spool **112** and the head housing **111** will be driven by a motor shaft **122a** to rotate synchronously, so that at this time the length of the trimming line **15** wound on the spool **112** extending from the effective portion **15a** of the head housing **111** will not change. That is, the trimming line **15** cannot be released, and the trimming head **11** is in a normal working state at this time. When the linkage device **113** is in the second equilibrium state, a relative movement occurs between the spool **112** and the head housing **111**. In this way, the trimming line **15** is thrown out relative to the spool **112** under the action of its own centrifugal force, so that the trimming line **15** is released, and the length of the effective portion **15a** of the trimming line **15** will increase.

(22) As shown in FIGS. 3 to 4, in some examples, the linkage device **113** includes a transmission member **113a** that can move freely between the spool **112** and the head housing **111**. When the linkage device **113** is in the first equilibrium state, the transmission member **113a** rotates synchronously with the spool **112**. At this time, the transmission member **113a** is in a first position, and the transmission member **113a** also mates with the head housing **111** to drive the head housing **111** to rotate synchronously with the spool **112**, and the motor has a first rotation speed at this time; When the linkage device **113** is in the second equilibrium state, the transmission member **113a** rotates with the spool **112**, and at the same time, the transmission member **113a** is disengaged from the head housing **111**. At this moment, the transmission member **113a** is in a second position. The spool **112** and the head housing **111** can move relative to each other, and the motor **122** has a first rotation speed at this time. As an implementation method, when the transmission member **113a** is in the first position or the second position, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed of the motor **122** to the first rotation speed is greater than or equal to 0 and less than or equal to 0.15. As an implementation method, when the transmission member **113a** is in the first position or the second position, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed of the motor **122** to the first rotation speed is greater than or equal to 0 and less than or equal to 0.1. In fact, in a better state, when the transmission member **113a** is in the first position or the second position, the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed of the motor **122** to the first rotation speed is greater than or equal to 0 and less than or equal to 0.05, and the ratio of the absolute value of the difference between the first rotation speed and the second rotation speed of the motor **122** to the first rotation speed approaches zero infinitely.

Through such a setting, it is realized that the trimming head **11** has no differential speed change and the autonomous line-feeding, that is, the automatic line-feeding in the true sense. When the driving member **113a** of the trimming head **11** is displaced, or the trimming line **15** is released to a preset length due to abrasion in the process of autonomous line-feeding, it all generates a preset force. So when the trimming head **11** is under the influence of the load of the trimming line **15** and the relative movement of the internal parts of the trimming head **11**, the rotation speed of the trimming head **11** itself has a preset change, so that the trimming head **11** cannot always rotate with a theoretical absolute constant speed, which resulting in a second speed that is different from the first speed. Here, within the allowable range of the above deviation, it is assumed that the change from the first speed to the second speed of the trimming head **11** is still within the scope of the aforementioned uniform rotation, and the switch of the trimming head **11** between the autonomous

line-feeding state and the working state does not depend on the aforementioned changes in speed.

(23) Optionally, in the direction of a straight line **102** which is substantially perpendicular to the direction of the rotation axis **101** or obliquely intersecting the rotation axis **101**, the transmission member **113a** is arranged between the spool **112** and the head housing **111** along the direction of the straight line **102**. The motor shaft **122a** and the spool **112** are configured to rotate synchronously. The transmission member **113a** is provided with a transmission hole through which the spool **112** is sleeved and can rotate synchronously with the spool **112**. In this way, when the motor shaft **122a** drives the spool **112** to rotate, the transmission member **113a** can rotate with the spool **112** synchronously. In fact, a limiting portion **112b** is formed or coupled connected to with the spool **112**, and the limiting portion **112b** can limit the circumferential displacement of the transmission member **113a** around the direction of the rotation axis **101**. Exemplarily, the limiting portion **112b** is a pair of protrusions symmetrically arranged with respect to a symmetry plane passing through the straight line **102**. The transmission member **113a** can be limited to the limiting portion **112b** when it is sleeved to the spool **112** in the direction of the rotation axis **101**, and only radial displacement in the linear direction perpendicular to the rotation axis **101** can be produced. The transmission member **113a** is asymmetrical with respect to the symmetry plane passing through the rotation axis **101** in terms of mass distribution. Optionally, a center of mass G of the transmission member **113a** deviates from the symmetry plane. The center of mass G of the transmission member **113a** deviates from the rotation axis **101**, so that when the trimming head **11** is driven to rotate by the motor **122**, the transmission member **113a** is subjected to a centrifugal force that makes it to have a tendency to move from the first position to the second position. When rotating around the axis of rotation **101**, a resistance against centrifugal force is applied to the transmission member **113a** to prevent the transmission member from moving from the first position to the second position. A biasing element **113b** is also provided between the transmission member **113a** and the spool **112**, and the biasing element **113b** can be a coil spring.

(24) Optionally, a receiving groove **112c** is formed and extending on the spool **112**, and the coil spring is at least partially disposed in the receiving groove **112c**. One end of the coil spring abuts against the bottom of the receiving groove **112c**, and the other end abuts the transmission member **113a**. The center of mass of the transmission member **113a** and the coil spring are arranged on two sides of the rotation axis **101**. In addition, the limiting portion **112b** and the receiving groove **112c** mate to form a sliding rail **117** for the transmission member **113a** to slide. That is, the transmission member **113a** also forms a slidable connection with the spool **112**. When the transmission member **113a** rotates with the spool **112**, the transmission member **113a** will generate a centrifugal force, and the transmission member **113a** is also subjected to a biasing force of the biasing element **113b**. The biasing force and the centrifugal force are in opposite direction. The centrifugal force and the biasing force can both be in the direction of the first straight line **102**. The direction in which the transmission member **113a** and the spool **112** constituting the sliding may also be along the direction of the straight line **102**. The center of mass of the transmission member **113a** and the biasing element **113b** are respectively arranged on two sides of a plane passing through the rotation axis **101** and perpendicular to the first straight line **102**. In some examples, when the transmission member **113a** is on the movement in the direction of the first straight line **102** relative to the spool **112**, it is also affected by a friction force between the transmission member **113a** and the spool **112**. The friction force can effectively prevent the biasing element **113b** from overcoming the centrifugal force, and prevent the linkage device **113** from being overly flexible and causing the misfeeding of the line.

(25) As shown in FIGS. 3 to 7, in the present example, the head housing **111** forms the accommodation space **114** mentioned above around the rotation axis **101**, the spool **112** is at least partially disposed in the accommodation space **114** formed by the head housing **111**, and the transmission member **113a** is at least partially located in the accommodating space **11**. Exemplarily, an upper bottom **111b** is formed on the end of the head housing **111** near the motor **122** in the

direction of the rotation axis **101**. The upper bottom **111b** does not enclose the accommodating space **114**, and the accommodating space **114** is partially opened upward, so that the motor shaft **122a** passes through the upper bottom **111b** and extends into the accommodation space **114**, and the transmission member **113a** partially protrudes out of the head housing **111**. The head housing **111** also form a mating portion **111c** that can be driven by the transmission member **113a**, and the transmission member **113a** is formed with a driving portion **113c** that mates with the mating portion **111c**. When the driving portion **113c** is mated with the mating portion **111c**, a force can be transmitted between the transmission member **113a** and the head housing. Optionally, the mating portion **111c** includes a mating surface **111d** and a return surface **111e**, wherein the mating surface **111d** is used to contact the driving portion **113c** to apply a reacting force to the driving portion **113c**; and the return surface **111e** is used to contact the driving portion **113c**. When the driving portion **113c** mates with the mating surface **111d**, the trimming head **11** is in the working state; when the driving portion **113c** is mated with the return surface **111e**, the trimming head **11** is in an autonomous line-feeding state.

(26) As shown in FIGS. **8** to **10**, an inner ring and an outer ring are formed in the direction where the upper bottom **111b** extends toward the motor **122**. Both the inner ring and the outer ring are formed around the rotation axis **101**. The inner ring is formed with first teeth **111f**, and the first teeth **111f** are external teeth formed on the inner ring and protruding toward the outer ring. The outer ring is formed with second teeth **111g**, and the second teeth **111g** are internal tooth formed on the outer ring and protruding toward the inner ring. Wherein, the first teeth **111f** constitutes or forms the above mating surface **111d**, and the second teeth **111g** constitutes or forms the above return surface **111e**. Wherein the number of the first teeth **111f** is multiple, and the number of the second teeth **111g** is also multiple. In the circumferential direction around the rotation axis **101**, the first teeth **111f** and the second teeth **111g** are staggered from each other. That is, the first teeth **111f** and the second teeth **111g** are not arranged in a diameter direction perpendicular to the rotation axis **101** at the same time. The transmission member **113a** is formed with a driving portion **113c** that can mate with first teeth **111f** or second teeth **111g**. The driving portion **113c** on the transmission member **113a** can be configured as an engaging tooth that mates with the first teeth **111f** and the second teeth **111g**. In this example, the driving portion **113c** and the biasing element **113b** are arranged on two sides of the transmission member **113a** respectively. That is, the center of mass of the driving portion **113c** and the transmission member **113a** are arranged on the same side of the rotation axis **101**.

(27) As shown in FIG. **12**, when the driving portion **113c** of the transmission member **113a** is engaged with the plurality of first teeth **111f** or the plurality of second teeth **111g** and the motor shaft **122a** rotates in the first rotation direction, that is, when the motor shaft **122a** rotates clockwise, the transmission member **113a** makes the head housing **111** to be relative static to the spool **112**, that is, there is no relative movement between the spool **112** and the head housing **111** at this time. It can be considered that the linkage device **113** is in the first equilibrium state at this time, and the string trimmer **100** is in the working state of normal grass trimming. As shown in FIG. **12**, when the driving portion **113c** of the transmission member **113a** is separated from the first teeth **111f** and the second teeth **111g**, the whole composed of the transmission member **113a** and the spool **112** can move relative to the head housing **111**. In this example, the direction of the movement of the spool **112** relative to the head housing **111** is consistent with the direction in which the string trimmer **100** trims grass. That is, the direction of the movement of the spool **112** relative to the head housing **111** is clockwise, that is, the spool **112** can now produce a relative movement relative to the head housing **111**, it can be considered that the linkage device **113** at this time is in the second equilibrium state, that is, the cutting head **11** is in the autonomously line-feeding state.

(28) In this example, a damping device **17** is also provided, and the damping device **17** has a damping spool **112** or head housing **111** to rotate and enables the relative rotation between the spool

112 and the head housing **111**. Exemplarily, the damping device **17** may be set as a limiting pin for restricting the rotation of the head housing **111** or the spool **112**, which can be inserted into a limiting hole reserved on the head housing **111** or the spool **112** to limit the rotation of the head housing **111** or the spool **112**. As another implementation, the damping device **17** can also be configured as a friction element which can contact the spool **112** or the head housing **111**, so that a relative motion between the head housing **111** and the spool **112** is caused by the speed difference. As another implementation, the damping device **17** can also be configured to include a one-way bearing **171** and a rotating support **172**; wherein the one-way bearing **171** can make the two parts or parts connected to it rotate relatively only in one rotation direction, but cannot rotate relatively in another rotation direction. The function of the rotating support **172** is to rotatably connect a portion of the cutting head **11** and form a support for its rotation. The rotating support can be the first housing **121** of the string trimmer **100** that houses the motor **122**, and it can also be other components fixedly coupled with the housing **121**, such as the guard **16**. Optionally, the one-way bearing **171** is arranged between the rotating support **172** and the head housing **111** so that they can form a one-way rotating connection. That is, when the rotating support **172** is used as a reference substance, the head housing **111** can rotate in one of the rotation directions, but cannot rotate in another direction. The motor shaft **122a** is connected to the spool **112** in a non-rotational manner so that the spool **112** can rotate in both directions relative to the rotating support. That is, when the rotating support **172** is used as the reference substance, the spool **112** can rotate forward and reverse.

(29) Based on the above hardware, when the motor **122** rotates forward, the motor shaft **122a** drives the spool **112** to rotate clockwise. At this time, the transmission member arranged on the spool **112** engages with the first teeth **111f** to achieve torque transmission, and at the same time, a one-way circumferential drive is set to make the head housing **111** rotate relative to the supporting member **172** (that is relative to the entire string trimmer **100**) and be able to rotate forward. Then, at this time, the spool **112** and the head housing **111** rotate synchronously, and the string trimmer **100** can execute the trimming mode. When the motor **122** rotates reversely, the motor shaft **122a** drives the spool **112** to reverse. The head housing **111** make the spool **112** and the head housing **111** to move relative to each other because the one-way bearing **171** prevents the head housing **111** from reversing. At this time, the driving portion **113c** on the transmission member is disengaged from the first teeth **111f** due to the reverse rotation of the motor **122**, so that the string trimmer **100** can perform an autonomous line-feeding mode.

(30) Exemplarily, as shown in FIGS. **8** to **9**, the tooth surface of the first teeth **111f** obliquely intersects the straight line **102** and forms a preset angle. The first teeth **111f** incline toward the first rotation direction. In some examples, the angle between the tooth surface of the first teeth **111f** and the straight line **102** is greater than 0° and less than or equal to 45° . During the grass trimming process of the string trimmer **100**, the trimming head **11** always keeps rotating at a constant speed. When the transmission member **113a** rotates with the spool **112** at the constant speed, there will be an interaction force F between the driving portion **113c** and the mating surface **111d** of the head housing **111**. Exemplarily, the force F is distributed along a direction perpendicular to mating surface **111d**. In this example, the mating surface **111d** extends in a first plane **103**, and the first plane **103** is substantially parallel to the rotation axis **101**. There is a preset minimum distance L between the rotation axis **101** and the first plane **103**, and the distance L is greater than zero. With this arrangement, the aforementioned interaction force F can be generated between the mating surface **111d** and the driving portion **1113**. When the lowest distance between the rotation axis and the first plane **103** is greater than 0 and less than or equal to 20 mm , the above-mentioned interaction force F will have better effect.

(31) In addition, the mating surface **111d** is not limited to the tooth surface which is smooth and continuous formed on the first teeth **111f** in this example. The mating surface **111d** can exist in any form as long as it can provide a force acting on the driving portion **113c**. In fact, the mating surface

111d includes an effective surface capable of providing a reaction force with the driving portion **113c**, and the effective surface has a projection line in a plane perpendicular to the rotation axis. The projection line includes a first extreme position **111da** and a second extreme position **111db** that can generate an interaction force with the driving portion **113c**. Here, it is defined that the effective acting surface is located between the first extreme position **111da** and the second extreme position **111db**, and the part beyond the first extreme position **111da** and the second extreme position **111db** cannot produce a reaction force with the driving portion **113c**. In fact, there may also be a section between the first extreme position **111da** and the second extreme position **111db** that cannot generate a reaction force with the driving portion **113c**. The section between the extreme position **111db** and the section where the reaction force can be generated between the driving portion **113c** is defined as the effective action surface. In this example, a central angle α is formed between the first extreme position **111da** and the second extreme position **111db** and the axis of the rotation axis, and the center angle α is greater than 0° and less than or equal to 45° . Within the range of the central angle α , the reaction force can be effectively generated between the mating surface **111d** and the driving portion **113c**. When the aforementioned central angle α is greater than or equal to 3° and less than or equal to 40° , the effect of the reaction force generated between the mating surface **111d** and the driving portion **113c** is better.

(32) However, during the rotation of the trimming head **11**, due to the change in the length of the trimming line **15**, the torque outputted by the trimming head **11** also changes accordingly. Exemplarily, the longer the trimming line **15** is, the greater the torque outputted by the trimming head **11** will be, and the greater the interaction force between the transmission member **113a** and the head housing **111** will be. On the contrary, the shorter the trimming line **15** is, the smaller the torque outputted by the head **11** will be, and the smaller the interaction force between the transmission member **113a** and the head housing **111** will be.

(33) Here, any point on the mating surface **111d** can be used for force analysis. As shown in FIG. **11**, the interaction force F between the first teeth **111f** of the head housing **111** and the driving portion **113c** of the transmission member **113a** is distributed along a direction of the tooth surface perpendicular to the mating surface **111d**, which can be decomposed into a first component force in the direction of the straight line **102** and a second component force F_1 in the direction of the straight line **102**, wherein the direction of the second component force F_1 is opposite to the direction of the centrifugal force F_0 . In addition, a friction is generated between the transmission member **113a** and the spool **112** due to the centrifugal force F_0 , and the friction is actually generated when the transmission member **113a** slides in the sliding rail **117** formed by the limiting portion **112b** and the receiving groove **112c**. Since the sliding rail **117** is distributed symmetrically with respect to a plane passing through the straight line **102** and the rotation axis **101** at the same time, that is, in the direction of a straight line perpendicular to the rotation axis **101**, the transmission member **113a** and the limiting portion **112b** include at least two contact surfaces **115** (see FIGS. **6** and **7**). Therefore, the friction force includes F_2 and F_3 , and F_2 and F_3 form a resultant force F_4 distributed along the direction of the straight line **102**. In addition, the biasing force between the transmission member **113a** and the biasing element **113b** can be defined as F_5 , and exemplarily, F_1 , F_4 , F_5 are all distributed along the direction of the straight line **102** and deviate from the centrifugal force F_0 . In this example, F_0 is only related to the mass of the transmission member **113a**, the angular velocity of the trimming head **11** and the radius of the spool **112**. That is, when the aforementioned parameters are all fixed values, F_0 is a constant. F_5 is the biasing force between the transmission member **113a** and the biasing element **113b**. When the elastic coefficient of the biasing element **113b** is determined, F_5 is also a constant. Therefore, during the rotation of the cutting head **11**, when the load of the trimming head **11** changes due to the change in the length of the trimming line **15**, only F changes, that is, F_1 changes. F_4 is a sliding friction force between transmission member **113a** and the spool **112**. When the material and contacting area of the transmission member **113a** and the spool **112** are determined, F_4 changes

synchronously with F1. Therefore, when the trimming line **15** is greater than a preset length, the torque of the trimming head **11** is relative large at this time, resulting in a large F1, so that $F_0 < F_1 + F_4 + F_5$, that is, the centrifugal force at this time is less than the summation of F1, F4 and F5. The transmission member **113a** cannot be separated from the first teeth **111f** under the action of centrifugal force, and the entire trimming head **11** is in the first equilibrium state. When the trimming line **15** is less than a preset length, the torque of the trimming head **11** is smaller at this time, resulting in a smaller F1, so that $F_0 > F_1 + F_4 + F_5$, that is, the centrifugal force at this time is greater than the summation of F1, F4 and F5. The transmission member **113a** is separated from the first teeth **111f** under the action of centrifugal force, and the entire trimming head **11** is in the second equilibrium state. At this time, the spool **112** can rotate relative to the head housing **111**, and the trimming head **11** starts to feed the trimming line.

(34) In one example, the angle between the tooth surface of the mating surface **111d** and the direction of the straight line **102** is 18° . When the length of the effective portion of the trimming line **15** is greater than a preset value, the torque at the output end of the motor is 0.505 N.Math.m. At this time, the mechanics calculation is carried out according to the principle of force reaction:
 $T = F * R$ (1)

(35) Where T represents the torque at the output end of the motor, F represents the interaction force between the spool **112** and the transmission member **113a** on a tooth surface perpendicular to the first teeth **111f**, and R represents a force radius of an action point on the tooth surface.

(36) It is deduced that when the length of the effective portion of the trimming line **15** is greater than the preset value, the force of the transmission member **113a** is as follows:

$$F = T/R = 0.505/37.5 * 1000 = 13.47N \quad (2)$$

$$F_1 = F * \sin 18 = 13.47 * \sin 18 = 4.16N \quad (3)$$

$$F_4 = F * k = 13.47 * 0.25 = 3.37N \quad (4)$$

$$F_5 = 3N \quad (5)$$

$$F_0 = mw\{\text{circumflex over ()}\}^2 * r = 0.0167 * 607.37 * 607.37 * 0.0014 = 8.62N. \quad (6)$$

(37) It can be concluded that when the length of the effective portion of the trimming line **15** is greater than the preset value, the relationship between the centrifugal force received by the transmission member **113a** and other resultant forces is $8.62 < 4.16 + 3.37 + 3 = 10.53$ and $F_0 < F_1 + F_4 + F_5$.

(38) Therefore, at this time, the centrifugal force received by the transmission member **113a** is smaller than the resultant force which driving the transmission member **113a** toward the driving portion **113c** to engage with the first teeth **111f** on the head housing **111**, and the transmission member **113a** keeps engaging with the head housing **111** when receiving the resultant force of all the forces.

(39) When the effective portion of the trimming line **15** is worn and shortened by 40 mm, the torque at the output end of the motor is 0.265 N.Math.m. At this time, it is deduced that when the length of the effective portion of the trimming line **15** is worn to less than the preset value, the force of the transmission member is as follows:

$$F = T/R = 0.24/37.5 * 1000 = 7.07N \quad (7)$$

$$F_1 = F * \sin 18 = 7.07 * \sin 18 = 2.18N \quad (8)$$

$$F_4 = F * k = 7.07 * 0.25 = 1.77N \quad (9)$$

$$F_5 = 3N \quad (10)$$

$$F_0 = mw\{\text{circumflex over ()}\}^2 * r = 0.0167 * 607.37 * 607.37 * 0.0014 = 8.62N \quad (11)$$

(40) It can be concluded that when the length of the effective portion of the trimming line **15** is less than the preset value, the relationship between the centrifugal force received by the transmission member **113a** and other resultant forces is $8.62 > 2.18 + 1.77 + 3 = 6.95$ and $F_0 > F_1 + F_4 + F_5$.

(41) Therefore, at this time, the centrifugal force received by the transmission member **113a** is greater than the resultant force of the engagement between the driving portion **113c** which drives the transmission member **113a** and the first teeth **111f** on the head housing **111**. Therefore, the

transmission member **113a** is under the resultant force of all the forces received and moves along the direction of the straight line to separate from the head housing **111**, so that the transmission member **113a** allows the head housing **111** and the spool **112** to rotate relative to each other at this time, and then the trimming line **15** can be released at this time.

(42) The mechanical calculation process shown above is only to facilitate the understanding of the force relationship between the spool **112**, the head housing **111** and the transmission member **113a** when the trimming head **11** is in the first equilibrium state and the string trimmer **11** rotates is at a constant speed. During use, the above data will have a preset deviation, which does not limit the protection scope of the present disclosure in other examples.

(43) As shown in FIGS. **12** to **13**, when the trimming head **11** is in the second equilibrium state, the transmission member **113a** is not constrained by the first teeth **111f** and the second teeth **111g**, and can continue to follow the direction of the straight line **102** under the action of centrifugal force and move toward the second teeth **111g**. When the driving portion **113c** of the transmission member **113a** moves to the second teeth **111g** of the head housing **111**, the driving portion **113c** starts to contact the return surface **111e**. During the relative rotation of the return surface **111e** between the head housing **111** and the spool **112**, the transmission member **113a** can be pushed to move along the straight line **102** toward the first teeth **111f**, and finally engaged with the first teeth **111f**, so as to complete a line-feeding and enter the next autonomously identification. In this example, the second teeth **111d** are not uniformly distributed in the circumferential direction around the rotation axis **101**, and they have gradual arc on the tooth surfaces facing the direction of the rotation axis **101**. The protrusions **111e** have a substantially continuous smooth surface to continuously push the transmission member **113a** back to the first teeth **111f** and engage with the first teeth **111f**. In fact, as an implementation, in a plane perpendicular to the rotation axis **101**, the central angle β formed by the line connecting a tooth tip or a tooth bottom of the second teeth **111g** together with the rotation axis **101** respectively is greater than or equal to 30° and less than or equal to 50° . The tooth surface of the second teeth **111g** close to the rotation axis **101** also includes a first section **111i** and a second section **111j**. Wherein, when the driving portion **113c** of the transmission member **113a** is located in the area where the first section **111i** is located, the first section **111i** does not generate the driving force on the transmission member **113a**. When the transmission member **113a** is separated from the first teeth **111f**, since the head housing **111** is not driven at this time, the head housing **111** is in a stalled state, so that it rotates relative to the spool **112**. At this time, the driving portion **113c** quickly moves to the second section **111j** and contacts the second section **111j**. During this process, the trimming line **15** is continuously released, and the return surface **111e** continuously pushes the driving portion **113c** to reset back to the first teeth **111f**. Exemplarily, in order for that the return surface **111e** can push the driving portion **113c** to reset to the first teeth, the return surface **111e** has a projection line in a plane perpendicular to the rotation axis **101**, and the return surface **111e** is configured to include a first extreme position **111ea** and a second extreme position **111eb** within the area of the projection line, and the first extreme position **111ea** and the second extreme position **111eb** can push the transmission member **113a** to return to the mating surface. The return surface **111e** is also configured to include a first circumference centered on the axis of the rotation axis **101** and passing through the first extreme position **111ea**, and a second circumference centered on the axis of the rotation axis **101** and passing through the second extreme position **111eb**. A angle γ formed by the tangent line of the first circumference at the first extreme position **111ea** and the tangent line of the second circumference at the second extreme position **111eb** is greater than or equal to 15° . In fact, when the driving portion **113c** is in contact with the return surface **111e**, when the driving portion **113c** is in the first extreme position **111ea**, at this time, the head housing **111** has just entered the stall state, and there is still a small speed difference between the driving portion **113c** and the spool **112**. When the driving portion **113c** is in contact with the return surface **111e**, the return surface **111e** has a first interaction force to push the driving portion **113c** to reset back to original place. As the relative rotation between the head housing **111** and the spool **112** continues,

the driving portion **113c** and the direct contact of the return surface **111e** is closer. At the same time, the relative speed between the head housing **111** and the spool **112** gradually increases, and the force between the return surface **111e** and the driving portion **113c** gradually increases until the driving portion **113c** is in the second position. At the second extreme position **111eb**, the return surface **111e** pushes the transmission member **113a** to completely recover. At this time, the return surface **111e** has a second interaction force that pushes the driving portion **113c** to recover, wherein the first interaction force is smaller than or equal to the second interaction force. Here, in order to enable the transmission member to be pushed and reset to the first teeth, the aforementioned angle γ is configured to be greater than or equal to 15° , so that the return surface has an effective length that can push the transmission member to the first position. Exemplarily, when the angle γ is configured to be greater than or equal to 20° , the return surface **111e** can be more reliably to push the transmission member **113a** to the first position, and at this time, the line-feeding effect of the trimming head **11** is better. In one example, if the return surface **111e** pushes the transmission member **113a** to the first position quickly, the trimming line **15** may not be released in time and effectively.

(44) In fact, the second section and the tooth surface of the first teeth **111f** away from the rotation axis **101** forms a guiding channel **111h**. The guiding channel can guide the driving portion **113c** to reset to the first position, that is, the position of the engagement of the driving portion **113c** and the first teeth **111f**. Exemplarily, due to the existence of the return surface **111e** and the continuous extension of the return surface **111e**, the driving portion **113c** will never engage with the second teeth **111g**. Thus, it is ensured that the length in every release cycle of the trimming line **15** is limited to a small range, so as to avoid the excessive release of the trimming line **15** which can lead to increase of the load of the motor **122** or avoid the trimming line **15** from breaking the guard **16** due to the excessively long length of the trimming line **15** being released.

(45) In fact, when turning on the tool, when the trimming head **11** accelerates from a static state to a constant speed state, if the trimming line **15** is greater than a preset length, the trimming line **15** will not be released due to the speed change; if the trimming line **15** is less than a preset length, the trimming line **15** will be released to a length greater than the preset length as the speed changes. In one example, the release of the trimming line **15** is not related to the speed change. It is only because the trimming line **15** is relatively short and the torque at the output end of the motor is relatively small, so that the trimming line **15** is released.

Claims

1. A string trimmer, comprising: a trimming head comprising: a spool for winding a trimming line, the spool comprising an outer surface having a limiting portion extending upwards from the outer surface, the limiting portion defining a channel; a head housing defining an accommodating space for at least partially accommodating the spool, the head housing comprising an outer housing surface defining an aperture through which the limiting portion extends, a circumference of the aperture including a plurality of mating portions extending from the outer housing surface, the plurality of mating portions including a driving portion having an inclined surface and a plane defined by the inclined surface; a transmission member slidably coupled to the spool within the channel, the transmission member comprising the driving portion, the transmission member having a first position where the driving portion mates with at least one of the plurality of mating portions of the head housing to prevent relative rotation between the spool and the head housing and a second position where the driving portion does not mate with at least one of the plurality of mating portions of the head housing to allow relative rotation between the spool and the head housing; a biasing element for biasing the transmission member towards the first position; and a driving device comprising a motor for driving the trimming head to rotate around a rotation axis, and the plane defined by the inclined surface does not pass through the rotation axis, wherein when the

transmission member is in the first position, the trimming head is in a working state and the transmission member rotates around the rotation axis synchronously with the spool, and when the transmission member is in the second position, the trimming head is in an autonomous line-feeding state and relative rotation is generated between the spool and the head housing to feed the trimming line, and when the trimming head is driven to rotate by the motor, the transmission member is subjected to a centrifugal force which causes a moving tendency in a direction from the first position to the second position, when the head housing rotates around the rotation axis, friction generated between the transmission member and the limiting portion applies a resistance to the transmission member against the centrifugal force to prevent the transmission member from moving from the first position to the second position, and when the trimming head is in a trimming state and a length of an effective portion of the trimming line extending beyond the head housing is less than a preset value, the resistance applied by the head housing to the transmission member is reduced so that the transmission member moves from the first position to the second position under the centrifugal force.

2. The string trimmer according to claim 1, wherein, when the length of the effective portion of the trimming line is greater than the preset value, the transmission member is in the first position and, when the length of the effective portion of the trimming line is less than the preset value, the transmission member is in the second position.

3. The string trimmer according to claim 1, wherein the limiting portion is defined by two upstanding walls symmetrically arranged with respect to the rotation axis, the limiting portion is formed with a sliding rail for the transmission member to slide, and the limiting portion comprises a contact surface in a straight line perpendicular to the rotation axis.

4. The string trimmer according to claim 1, wherein the spool further comprises a receiving groove, the biasing element is arranged in the receiving groove, a first end of the biasing element abuts against the receiving groove, and a second end of the biasing element abuts against the transmission member.

5. The string trimmer according to claim 1, wherein the trimming head further comprises an automatic winding state, when the trimming head is in the automatic winding state, the motor drives the spool to rotate relative to the head housing in a second rotation direction so that the trimming line is wound to the spool and, when the trimming head is in the autonomous line-feeding state, the transmission member moves to the second position and the spool rotates relative to the head housing along a first rotating direction to release the trimming line.

6. The string trimmer according to claim 1, wherein the string trimmer further comprises a damping device for damping at least one of the spool or the head housing to make the string trimmer in an automatic winding state.

7. The string trimmer according to claim 1, wherein the plane defined by the inclined surface is parallel to the rotation axis, and a distance between the rotation axis and the plane is greater than zero.

8. The string trimmer according to claim 1, wherein a distance between the rotation axis and the plane defined by the inclined surface is greater than 0 mm and less than or equal to 20 mm.

9. The string trimmer according to claim 7, wherein the inclined surface is capable of applying a reaction force, the inclined surface has a projection line in the plane defined by the inclined surface and perpendicular to the rotation axis, the projection line comprises a first extreme position and a second extreme position, and a central angle between a connecting line between the first extreme position and the rotation axis and the connecting line between the second extreme position and the rotation axis is greater than 0° and less than or equal to 45° .

10. The string trimmer according to claim 9, wherein the string trimmer further comprises a first circumference centered on the rotation axis and passing through the first extreme position and a second circumference centered on the rotation axis and passing through the second extreme position, an angle γ formed by a tangent line of the first circumference at a first extreme position,

and a tangent line of the second circumference at the second extreme position is less than or equal to 45° .

11. The string trimmer according to claim 1, wherein, when the transmission member is in the first position, the trimming head is in the trimming state and the motor has a first rotation speed and, when the transmission member is in the second position, the trimming head is in the autonomous line-feeding state, the motor has a second rotation speed, a ratio of an absolute value of a difference between the first rotation speed and the second rotation speed to the first rotation speed is greater than or equal to 0 and less than or equal to 0.1, and the relative rotation is generated between the spool and the head housing to feed the trimming line.

12. A trimming head, used for a string trimmer, capable of being driven to rotate around a rotation axis to achieve grass trimming, comprising: a spool for winding a trimming line the spool comprising an outer spool surface having a limiting portion extending upwards from the outer spool surface, the limiting portion defining a channel; a head housing forming an accommodating space for at least partially accommodating the spool, the head housing comprising an outer head housing surface defining an aperture through which the limiting portion of the spool extends, a circumference of the aperture including a plurality of mating portions extending from the outer head housing surface, the plurality of mating portions including a driving portion having an inclined surface and a plane defined by the inclined surface; and a transmission member slidably coupled to the spool within the channel, the transmission member having a first position where the driving portion mates with at least one of the plurality of mating portions of the head housing to prevent relative rotation between the spool and the head housing and a second position where the driving portion does not mate with at least one of the plurality of mating portions of the head housing to allow relative rotation between the spool and the head housing; wherein the transmission member rotates around the rotation axis synchronously with the spool and the trimming head is in a trimming state when the transmission member is in the first position, the rotation is generated between the spool and the head housing to feed the trimming line when the transmission member is in the second position, a center of gravity of the transmission member deviates from the rotation axis, when the trimming head is rotating, the transmission member is subjected to a centrifugal force which cause a moving tendency in a direction from the first position to the second position, when the head housing rotates around the rotation axis, the head housing also applies a resistance to the transmission member against the centrifugal force to prevent the transmission member from moving from the first position to the second position, and when the trimming head is in the trimming state and a length of an effective portion of the trimming line extending beyond the head housing is less than a preset value, the resistance applied by the head housing to the transmission member is reduced so that the transmission member moves from the first position to the second position under the centrifugal force.

13. The string trimmer according to claim 12, wherein the transmission member comprises the driving portion for driving the head housing to rotate synchronously with the spool, the head housing is formed with a mating portion for mating with the driving portion, the mating portion comprises first teeth distributed around the rotation axis, the head housing is formed with or connected with second teeth for resetting the transmission member, the first teeth and the second teeth are staggered around the rotation axis, in a circumferential range around the rotation axis, each of the second teeth has a tooth surface distributed toward the rotation axis, and the tooth surface has a return surface facing the rotation axis and the return surface is gradually changing.

14. A trimming head, used for string trimmer, capable of being driven to rotate around a rotation axis to achieve grass trimming, comprising: a spool for winding a trimming line the spool comprising an outer surface having a limiting portion extending upwards from the outer surface, the limiting portion defining a channel; a head housing formed with an accommodating space for at least partially accommodating the spool the head housing comprising an outer housing surface defining an aperture through which the limiting portion extends, a circumference of the aperture

including a plurality of mating portions extending from the outer housing surface, the plurality of mating portions including a driving portion having an inclined surface and a plane defined by the inclined surface; and a transmission member slidably mounted to the spool within the channel, the transmission member comprising the driving portion, the transmission member having a first position where the driving portion mates with at least one of the plurality of mating portions of the head housing to prevent relative rotation between the spool and the head housing and a second position where the driving portion does not mate with at least one of the plurality of mating portions of the head housing to allow relative rotation between the spool and the head housing; wherein the transmission member drives the head housing to rotate synchronously with the spool, the head housing is formed with a mating portion to mate with the driving portion, the mating portion is formed with a mating surface for contacting with the driving portion to apply a reacting force to the driving portion, the mating surface extends in a first plane, the first plane is parallel to the rotation axis, and a distance between the rotation axis and the first plane is greater than zero.

15. The trimming head according to claim 14, wherein the distance between the rotation axis and the first plane is greater than 0 and less than or equal to 20 mm.

16. The trimming head according to claim 14, wherein the inclined surface is capable of applying a reaction force, the inclined surface having a projection line in a plane perpendicular to the rotation axis, the projection line comprises a first extreme position and a second extreme position, and a central angle between a connecting line between the first extreme position and the rotation axis and the connecting line between the second extreme position and the rotation axis is greater than 0° and less than or equal to 45° .
